RAJALAKSHMI ENGINEERING COLLEGE

Department of Artificial Intelligence and Machine Learning

AD23632 – Framework for Data and Visual Analytics (FDVA)

MINI PROJECT REPORT

IPL CRICKET ANALYTICS

MATCH INSIGHTS – TEAM PERFORMANCE

Submitted in partial fulfilment of the requirements for the course Framework for Data and Visual Analytics

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Note: Page numbers may differ from the reference report because this document is formatted around the uploaded Power BI project.

ABSTRACT

The project titled **IPL Cricket Analytics – Match Insights and Team Performance** uses Microsoft Power BI to present cricket match data through visualizations. The uploaded Power BI report contains a dashboard page with charts configured around the *matches* data table. The visual definitions include a clustered bar chart comparing winners, a donut chart for winners, a pie chart for toss decisions, a line chart using the date hierarchy, a gauge based on toss-winner records, and a funnel chart by season. The dashboard heading reads "IPL Cricket Analytics – Match Insights: Team Performance."

The purpose of the project is to make match-level information easier to explore by presenting team outcomes, toss decisions, and season-wise match patterns visually. Power BI allows the viewer to compare categories and identify patterns without manually reviewing each record. Exact KPI totals and chart values are not stated in this report because they could not be reliably read from the report definition alone; the results section therefore focuses on the visual structure and the kinds of comparisons supported by the dashboard.

Keywords: IPL Cricket Analytics, Microsoft Power BI, Data Visualization, Match Analysis, Team Performance, Toss Decision, Season Analysis.

CHAPTER 1: INTRODUCTION

1.1 Introduction

The Indian Premier League (IPL) is a professional cricket tournament in which match outcomes, team performance, toss decisions, and season-level patterns are useful areas of analysis. Match datasets contain structured information that can be summarized to compare teams and explore changes across seasons. A dashboard presents these details in a visual form, making it easier to identify patterns and ask follow-up questions.

This project uses Microsoft Power BI to build an IPL Cricket Analytics dashboard. The report heading identifies its focus as match insights and team performance. Its visual configuration includes winner-based comparisons, toss-decision distribution, a date-based line chart, a gauge, and a season-based funnel chart.

1.2 Problem Statement

Match records are often stored in tables that require users to sort, group, and compare records manually. Without visual summaries, it may take more time to examine how match winners, toss decisions, and seasons are represented in the dataset. A dashboard is therefore useful for bringing selected match attributes together in a single visual interface.

1.3 Objectives

• To create an IPL cricket analytics dashboard using Microsoft Power BI.

• To visualize match records grouped by the winning team.

• To present the distribution of toss decisions.

• To explore date-based patterns in match records.

• To include season-wise visual comparison.

• To make match data easier to interpret through charts.

1.4 Scope of the Project

The scope is limited to the match-related fields configured in the uploaded Power BI report, including winner, toss decision, date, season, toss winner, and match ID. The dashboard supports descriptive exploration of the available dataset. It does not, by itself, establish the causes of a match outcome or predict future results.

CHAPTER 2: TOOLS AND TECHNOLOGIES USED

2.1 Microsoft Power BI

Microsoft Power BI is a business intelligence platform for connecting to data, modeling information, and creating interactive reports. In this project, it is used to configure visualizations around fields in the matches table. The supplied file contains a report definition with a single page named “Page 1” and six chart visuals plus a text heading.

2.2 Dataset and Data Model

The report definition references a table named *matches* and a second reference named *matches (2)* for one gauge visual. The file also contains a Power BI data model. This report describes only fields visible in the visualization definitions; a complete data dictionary and exact record counts should be confirmed from the source dataset or Power BI Data view.

2.3 Visualization Types

Visualization	Configured purpose / fields
Clustered bar chart	Category: winner, Y: id
Pie chart	Category: toss_decision, Y: id
Line chart	Category: Year, Y: id
Donut chart	Category: winner, Y: id
Gauge	Y: toss_winner
Funnel chart	Category: season, Y: id

2.4 Technology Summary

The confirmed primary tool is Microsoft Power BI Desktop, represented by the uploaded .pbix file. The report does not provide evidence that Python, Pandas, NumPy, or Matplotlib were used; therefore, they are not listed as implemented technologies in this report.

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

The Power BI visual definitions refer to a match dataset with fields for match ID, winner, toss decision, date, season, and toss winner. These attributes support grouping and comparing records by match outcome, toss-related choices, and season. The exact source name, number of rows, date range, and full list of columns are not exposed in the visual definitions inspected for this report.

3.2 Dataset Attributes

Attribute	Description based on the report
id	Match identifier; used as an aggregated field in several visuals.
winner	Winning team; used as a category in the clustered bar and donut charts.
toss_decision	Decision made after winning the toss; used in the pie chart.
date	Match date; the line chart uses the Year level of the date hierarchy.
season	Season label; used as the category in the funnel chart.
toss_winner	Team that won the toss; used in the gauge aggregation.

3.3 Data Considerations

The report configuration aggregates the *id* field in several visuals. Since an identifier is not necessarily a measure of performance, the user should verify the aggregation and field selection in Power BI before interpreting chart heights as match counts. This report therefore describes the visual configuration rather than claiming precise numerical findings that are not verified.

CHAPTER 4: METHODOLOGY

4.1 Project Workflow

The project workflow represented by the supplied file can be summarized as:

Match Dataset / Data Model → Configure Fields → Create Power BI Visuals → Compare Categories → Interpret Match Patterns

4.2 Data Preparation

Before interpreting the dashboard, the source data should be checked for missing values, duplicate match identifiers, inconsistent team names, and valid date and season formats. The uploaded report file confirms the fields used in visuals, but it does not document a separate data-cleaning procedure. Any cleaning steps should therefore be recorded only if they were actually performed.

4.3 Visual Design

The dashboard page uses a heading across the top and positions the visuals in a two-row layout. The configured chart types are a clustered bar chart, pie chart, line chart, donut chart, gauge, and funnel chart. This combination provides categorical comparisons, distributions, a time-based view, and a gauge-style summary.

4.4 Interpretation

Interpretation should begin by confirming each visual's title, category, aggregation, and filter settings. A chart should be described according to what its axes and labels actually measure. For example, a chart grouped by winner can compare categories, but the selected measure must be confirmed before concluding that it represents the number of matches won.

CHAPTER 5: POWER BI DASHBOARD AND VISUALIZATIONS

5.1 Dashboard Overview

The uploaded Power BI report contains one page, displayed as “Page 1,” with the title “IPL Cricket Analytics – Match Insights: Team Performance.” The report layout is widescreen and includes six chart visuals beneath the heading.

5.2 Clustered Bar Chart: Winner Comparison

The clustered bar chart uses *winner* as its category and an aggregation of *id* as its value. It is configured to sort by the aggregated value in descending order. This visual can compare winner categories; however, its meaning as a count of wins should be verified by checking the aggregation or replacing it with a count of match IDs if appropriate.

5.3 Donut Chart: Winner Distribution

The donut chart uses *winner* as the category and an aggregation of *id* as the value. It provides a part-to-whole view of the displayed categories. The chart can help compare the relative share of each winner category once the measure is confirmed.

5.4 Pie Chart: Toss Decision

The pie chart uses *toss_decision* as the category and the aggregated *id* field as its value. It is intended to show the distribution between toss-decision categories in the dataset. The exact proportions are not reported here because numeric visual output was not available in the uploaded report definition.

5.5 Line Chart: Date-Based Pattern

The line chart uses the Year level of the date hierarchy on its category axis and an aggregation of *id* on the value axis. This makes it possible to compare the configured measure across years. Confirm the aggregation before interpreting the line as the number of matches played per year.

5.6 Gauge: Toss-Winner Records

The gauge visual uses a count of non-null *toss_winner* values and has a configured minimum, maximum, and target. It provides a single-value indicator. The configured target should be reviewed to ensure it is meaningful for the dataset and project objective.

5.7 Funnel Chart: Season Comparison

The funnel chart uses *season* as its category and an aggregation of *id* as its value. It can display a ranked or staged-looking comparison by season, although seasons are not inherently sequential funnel stages. A bar chart may be more suitable if the intended goal is simply to compare match counts by season.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results

The uploaded file confirms that the dashboard has been assembled with six visualizations and a title focused on IPL match insights and team performance. The visuals cover winner categories, toss decisions, date hierarchy, toss-winner records, and seasons. This gives the user multiple ways to explore the match dataset from a single page.

6.2 Discussion

The dashboard combines distribution charts with categorical and time-based visuals. The bar chart and donut chart both use winner as a category, while the pie chart examines toss decisions. The line chart groups the date hierarchy by year, and the funnel chart groups records by season. The gauge provides a count-based summary for the toss-winner field.

6.3 Validation Needed Before Final Submission

The report definition shows that several visuals aggregate the *id* field using a sum. Since match IDs are identifiers, a sum of IDs generally does not represent match frequency or team wins. Before presenting numerical conclusions, open the report in Power BI and check the Values field for these visuals. If the objective is to count matches, use a count of match ID or an appropriate row-count measure. Also confirm the gauge target and the meaning of the funnel chart.

No exact totals or rankings are claimed in this chapter because they cannot be verified from the visual configuration alone. Once the dashboard is checked in Power BI, actual values can be inserted into the final results.

CHAPTER 7: ADVANTAGES, LIMITATIONS AND FUTURE ENHANCEMENTS

7.1 Advantages

• Presents selected IPL match attributes in a single dashboard.

• Makes winner and toss-decision categories easier to compare.

• Includes a date-based visual for exploring changes across years.

• Uses multiple chart types to support different forms of comparison.

• Provides a foundation for further interactive analysis in Power BI.

7.2 Limitations

• The report contains one page, so all visuals share the same dashboard space.

• Exact numeric results and source dataset details are not documented in the report definition.

• Summing match IDs may not be an appropriate measure for match counts or team wins.

• The gauge target and funnel representation should be checked for analytical relevance.

• The dashboard describes the available data and does not establish causal relationships.

7.3 Future Enhancements

• Add KPI cards for total matches, seasons, teams, and venues if supported by the dataset.

• Use a count of match IDs for match-frequency analysis.

• Add slicers for season, team, venue, and toss decision where relevant fields exist.

• Add visuals for wins by team, toss outcome versus match outcome, and venue-wise performance.

• Use clearer titles, axis labels, and tooltips to explain each measure.

• Add a data-source and data-cleaning note to document dataset provenance.

CHAPTER 8: POWER BI DASHBOARD OUTPUT

The uploaded .pbix file was inspected for its report layout and visual definitions. The following inventory describes the elements found in the dashboard configuration. For the final submission, export a screenshot from Power BI Desktop and place it on this page if your faculty requires actual dashboard screenshots.

Dashboard element	Configured field(s)	Visual type
Clustered bar chart	Category: winner, Y: id	Clustered bar chart
Pie chart	Category: toss_decision, Y: id	Pie chart
Line chart	Category: Year, Y: id	Line chart
Donut chart	Category: winner, Y: id	Donut chart
Gauge	Y: toss_winner	Gauge
Funnel chart	Category: season, Y: id	Funnel chart
Dashboard heading	IPL Cricket Analytics – Match Insights: Team Performance	Text box

Screenshot insertion note: In Power BI Desktop, open the report page, set the zoom so the full dashboard is visible, capture a screenshot, and insert it below this inventory in the final Word/PDF version.

CHAPTER 9: CONCLUSION

The IPL Cricket Analytics mini project demonstrates how Microsoft Power BI can be used to present match-related data through a dashboard focused on match insights and team performance. The uploaded report includes visuals configured to compare winners, display toss-decision distribution, explore year-based patterns, summarize toss-winner records, and compare seasons.

The project provides a starting point for descriptive cricket analytics by bringing several views of the dataset together. To strengthen the analytical validity of the report, the field aggregations should be reviewed—especially the use of a sum of match IDs—and the chart measures should be aligned with the intended questions. After verifying these settings and capturing the final dashboard screenshot, the report can be updated with exact numerical results and observations.

Overall, the project provides practical experience in Power BI report construction, field selection, chart configuration, and data visualization. It shows how a structured match dataset can be organized into visual summaries that support exploration and interpretation.

— End of Report —